

CHAPTER 4

Unsaturated Flow

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Introduction

Water flow in unsaturated soils is a very important field of study. It is relevant to proper irrigation and drainage management, the study of contaminant transport in unsaturated soil, non-lithified geological materials and fractured rock, estimating soil water budgets, soil engineering, etc. Many soils around the world are unsaturated for at least part of the year. Therefore, it is imperative to understand the fundamental physics of water flow in unsaturated soils to tackle many of the world's ecological problems.

Unsaturated flow differs from saturated flow (Chapter 3) in that not all of the soil pores are filled with water; that is, soil gas occupies part of the pore space of unsaturated soils (we are neglecting other nonwetting phases, such as oil in a water-oil system). The fact that hydraulic conductivity is a function of the degree of saturation is the main difference between calculating flow in unsaturated and saturated soils. So, with unsaturated flow, we must consider both spatial and temporal variability in the hydraulic conductivity of the soil, whereas with saturated flow, generally only spatial variability need be considered. Fortunately, the 25 problems given below deal almost exclusively with homogeneous and isotropic soils under steady state flow conditions!

Infiltration of water at the soil surface is a very important component of unsaturated (or saturated) flow. The rate of infiltration during a rainfall or irrigation event will ultimately determine the amount of water that runs off the land surface during that event. Runoff can lead to soil erosion and contamination of nearby surface waters. It is, therefore, important to understand how to estimate infiltration rates and time to ponding of water on the land surface during a heavy rainfall or irrigation.

Conducting unsaturated-flow experiments under laboratory or field conditions requires the use of specialized equipment that can maintain a negative pressure head (below atmospheric) within a soil. In the laboratory, this may require the use of permeable membranes with a high air-entry pressure head attached to the ends of soil columns. A vacuum pump or hanging water table system may then be used to apply negative pressure head conditions within the soil. Under field conditions, irrigating the soil at a rate less than its saturated hydraulic conductivity or using a tension infiltrometer will produce unsaturated flow conditions. Another approach, known as the instantaneous profile method, measures unsaturated flow parameters under drainage from an initially saturated soil. The following problems deal with a variety of unsaturated flow conditions and assumptions – enjoy!

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Symbols and Notation

C	Specific water capacity	L^{-1}
D	Soil water diffusivity	$L^2 T^{-1}$
g	Gravity constant	$L T^{-2}$
G	Shape parameter (Reynolds and Elrick, 1990)	dimensionless
h	Soil-water pressure head	L
h_{ae}	Air entry pressure head	L
h_f	Wetting front pressure head	L
h_z	Gravitational head	L
H	Total hydraulic head	L
i	Infiltration rate	$L T^{-1}$
I	Cumulative infiltration	L
K	Hydraulic conductivity	$L T^{-1}$
K_{fs}	Field- saturated hydraulic conductivity	$L T^{-1}$
K_s	Saturated hydraulic conductivity	$L T^{-1}$
m	Shape parameter (van Genuchten, 1980)	dimensionless
n	Shape parameter (van Genuchten, 1980)	dimensionless
PAW	Plant Available Water	$L^3 L^{-3}$
q	Flux density	$L T^{-1}$
Q	Volumetric flow rate	$L^3 T^{-1}$
R	Rainfall rate	$L T^{-1}$
S	Sorptivity	$L T^{-1/2}$
t_p	Time to ponding	T
$\bar{V}$	Pore water velocity	$L T^{-1}$
V	Volume	L^3
α_{VG}	Shape parameter (van Genuchten, 1980)	L^{-1}
α_{BC}	Shape parameter (Brooks and Corey, 1964)	L^{-1}
α_G	Shape parameter (Gardner, 1958)	L^{-1}
θ	Water content (volumetric)	$L^3 L^{-3}$
θ_c	Water content at field capacity	$L^3 L^{-3}$
θ_i	Initial water content	$L^3 L^{-3}$
θ_{wp}	Water content at wilting point	$L^3 L^{-3}$
θ_r	Residual water content	$L^3 L^{-3}$
θ_s	Saturated water content	$L^3 L^{-3}$
λ	Shape parameter (Brooks and Corey, 1964)	dimensionless
ρ_b	Bulk density	ML^{-3}
ρ_s	Particle density	ML^{-3}
ϕ	Porosity	$cm^3 cm^{-3}$
ϕ_m	Matric flux potential	$L^2 T^{-1}$

Problem 4-1

A 50 cm long, vertical soil column with a cross-sectional area, A_s of 50 cm² has the lower end connected to a constant-head device maintained at a soil-water pressure head, h of -50 cm. Water is evaporating from the other end of the column at a steady volumetric flow rate, Q of 50 cm³ h⁻¹.

a. Plot the gravitational head (h_z), the pressure head (h), and the total hydraulic head (H) for this column. Explain why you plotted H and h as you did.

Answer:

In order for upward movement of water to occur, the pressure head (h) line must be in the shaded area, starting from a head value of -50 cm and decreasing linearly through the length of the column. Therefore, the hydraulic head (H) will decrease as you move up through the column. If the pressure head were along the dashed line, there would be no flow.

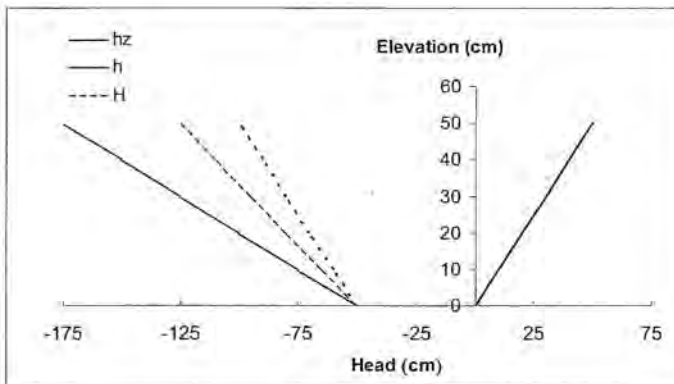


Fig. 4-1: Pressure head distribution

b. If the "average" hydraulic conductivity of the soil in the column is 0.5 cm h⁻¹, what is the soil-water pressure head (h) at the evaporating surface?

Assume z is positive upward. A positive q indicates that flow of water is in the upward direction

Answer:

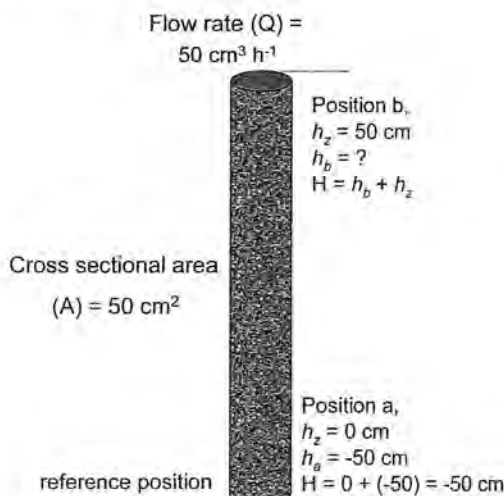


Fig. 4-2:
Soil column
under constant
evaporation rate

Start with the unsaturated form of Darcy's Equation

$$q = -K \frac{dH}{dz}$$

Also recall that,

$$q = \frac{Q}{A_y} = \frac{50 \text{ cm}^3 \text{ h}^{-1}}{50 \text{ cm}^2} = 1 \text{ cm h}^{-1}$$

Now substitute the solved value of the flux density (q) and the other known values from the question into Darcy's Equation, and assume that K is a constant.

$$1 = -0.5 \frac{(h_{posb} + 50) - (-50 + 0)}{50 - 0}$$

$$-100 = h_{posb} + 50 + 50$$

$$h_{posb} = -200 \text{ cm}$$

Problem 4-2

What would be the average evaporation rate (cm d^{-1}) over a 5-day period from a soil column with a saturated water content, θ_s of $0.46 \text{ cm}^3 \text{ cm}^{-3}$? The volumetric water content, θ on the fifth day is:

Table 4-1: Water content distribution after 5 days

z (cm)	θ (cm ³ cm ⁻³)
0-10	0.35
10-20	0.40
20-30	0.45
>30	0.46

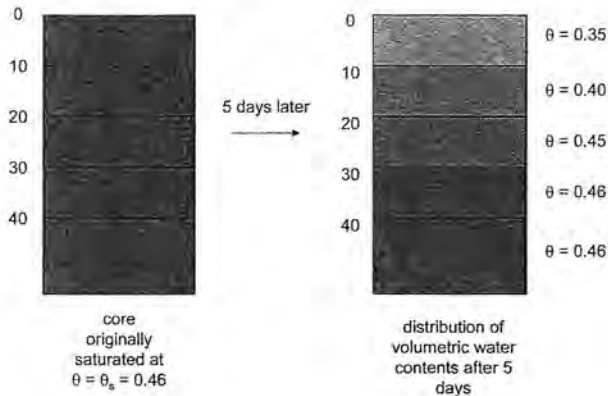


Fig. 4-3: Water content distributions before and after 5 days evaporation

Find the change in θ for each layer.

Answer:

$$\Delta\theta_{(0 \text{ cm} - 10 \text{ cm})} = (0.46 - 0.35) = 0.11 \text{ cm}^3 \text{ cm}^{-3}$$

$$\Delta\theta_{(10 \text{ cm} - 20 \text{ cm})} = (0.46 - 0.40) = 0.06 \text{ cm}^3 \text{ cm}^{-3}$$

$$\Delta\theta_{(20 \text{ cm} - 30 \text{ cm})} = (0.46 - 0.45) = 0.01 \text{ cm}^3 \text{ cm}^{-3}$$

$$\Delta\theta_{(> 30 \text{ cm})} = (0.46 - 0.46) = 0.0 \text{ cm}^3 \text{ cm}^{-3}$$

Determine the amount of water evaporated

$$\text{Water evaporated} = \sum \Delta\theta (\text{depth of each layer}) = (0.11 + 0.06 + 0.01 + 0.0)10 \text{ cm} = 1.8 \text{ cm}$$

Over the 5 days the average evaporation rate would be

$$\frac{1.8 \text{ cm}}{5 \text{ days}} = 0.36 \text{ cm d}^{-1}$$

Problem 4-3

a. A soil sample is placed in a Tempe pressure cell (see below), and is subjected to a positive gas pressure head of 500 cm. If the water in the soil sample is at hydraulic equilibrium (no flow), compute the corresponding soil water pressure head (h) in the center of the soil core. Show your calculations.

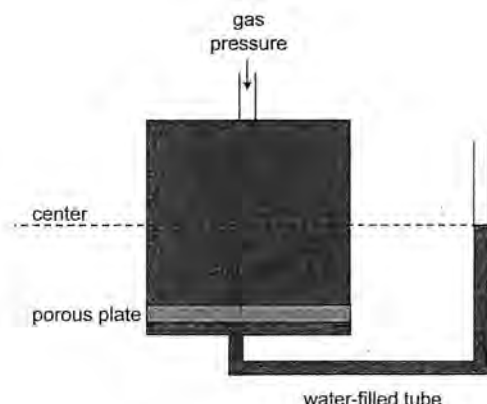


Fig. 4-4: Soil sample inside a Tempe pressure cell

Answer:

Assume the reference position for the gravitational head (h_z) is the bottom of the core. If there is no flow through the core, then the slope of the pressure head (h) must be equal to the negative slope of the gravitational head. Under no-flow conditions, the pressure head will increase through the core with an increase in unit head per decrease in unit gravitational head (starting from the top of the core).

$$H = h_z + h \quad (\text{for any position within the core})$$

$$d = \text{height of the core (cm)}$$

$$h_z = d, \quad \text{top of the core}$$

$$H = 500 + d, \quad \text{at the top of the core}$$

At the center of the core, H will still be equal to that at the top of the core :

$$H = h_{\text{middle of the core}} + h_{z, \text{middle of the core}}$$

$$500 + d = h_{\text{middle of the core}} + \frac{d}{2}$$

$$h_{\text{middle of the core}} = 500 + \frac{d}{2}$$

The pressure head at the center of the core is equal to the applied pressure head at the top, plus half the height of the core.

b. What is the maximum allowable radius of the pores in the porous plate so that its pores will not drain when the 500 cm of gas pressure head is applied to the soil core?

Answer:

Pore radius, $r = \frac{2(\cos \gamma)\sigma}{\rho_w g h}$ where γ is the contact angle between the

water and the side of the pore and σ is the surface tension. Assume the contact angle is 0, $\cos(0)=1$ and $\sigma = 72 \text{ dynes cm}^{-1}$ at 25°C (dyne = g cm s^{-2}), $g = 981 \text{ cm s}^{-2}$, $\rho_w = 1.0 \text{ g cm}^{-3}$:

$$r = \frac{2(72 \text{ g cm s}^{-2} \text{ cm}^{-1})}{1.0 \text{ g cm}^{-3} 981 \text{ cm s}^{-2} 500 \text{ cm}} = 2.94(10)^{-4} \text{ cm}$$

The maximum pore radius that will not drain at 500 cm applied pressure is $2.94(10)^{-4} \text{ cm}$.

Problem 4-4

a. In the soil column arrangement shown below, water flow is at steady state. The air-entry value, h_{ae} of the soil is -25 cm .

Is the pressure head, h :

- 1) Increasing from left to right?
- 2) Decreasing from left to right?
- 3) Constant along column?
- 4) Unknown?

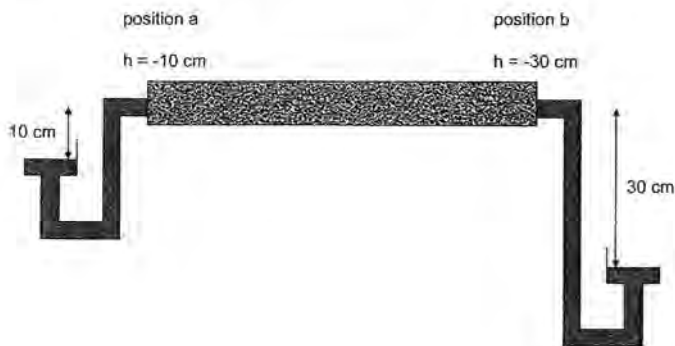


Fig. 4-5:
Horizontal soil
column under
steady state
water flow

Answer:

Since the column is horizontal, and saturated at the left side and unsaturated at the right side, the pressure head is decreasing from left to right.

b. After the depth of the hanging water column on the right-hand side of the column was reduced from 30 to 20 cm, the steady state flux density, q is 10 cm d^{-1} and the soil volumetric water content, θ is $0.4 \text{ cm}^3 \text{ cm}^{-3}$. What is the average pore water velocity, $\bar{v}$?

- 1) 1 cm d^{-1}
- 2) 25 cm d^{-1}
- 3) 4 cm d^{-1}
- 4) need more information

Answer:

$$\bar{v} = \frac{q}{\theta} = \frac{10 \text{ cm d}^{-1}}{0.4 \text{ cm}^3 \text{ cm}^{-3}} = 25 \text{ cm d}^{-1}$$

Problem 4-5

Water is flowing at a steady rate in a 50 cm long horizontal unsaturated soil column. The soil pressure head (h) is 0 cm at the right end and -180 cm at the left end of the soil column. Both heads are maintained throughout the steady state experiment. The saturated hydraulic conductivity, K_s is 4 cm d^{-1} . The unsaturated hydraulic conductivity of the soil at a pressure head of -180 cm is one tenth of K_s . The unsaturated hydraulic conductivity, $K(h)$, is given by $K(h) = A e^{\alpha_G h}$, where $h < 0$.

a. Calculate the coefficients A and α_G .

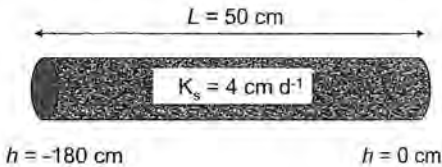


Fig. 4-6: Horizontal soil column under steady state water flow

Answer:

$$K(-180) = 0.4 \text{ cm d}^{-1}$$

$$K(h) = A e^{\alpha_G h}$$

$$A = K_s = 4 \text{ cm d}^{-1}$$

$$0.4 = 4e^{\alpha_G(-180)}$$

$$\ln 0.1 = \alpha_G(-180)$$

$$\alpha_G = \frac{-2.3}{-180 \text{ cm}} = 0.013 \text{ cm}^{-1}$$

b. Calculate the flux density. Assume $x = 0$ on the left and $x = L = 50 \text{ cm}$ on the right.

Answer:

$$q = -K(h) \frac{dh}{dx}$$

$$\int_0^{50} q \, dx = \int_{-180}^0 -K(h) \, dh$$

$$50q = - \int_{-180}^0 4e^{0.013h} \, dh$$

$$12.5q = - \int_{-180}^0 e^{0.013h} \, dh = - \frac{1}{0.013} e^{0.013h} \Big|_{-180}^0$$

$$-0.1625q = 1 - e^{0.013(-180)}$$

$$-0.1625q = 1 - 0.096$$

$$q = -5.56 \text{ cm d}^{-1}$$

The negative number is consistent with flow in the negative- x direction.

c. Find an analytical expression for h as a function of distance (x).

Answer:

$$q = -K(h) \frac{dh}{dx}$$

$$\int_0^x q \, dx = - \int_{-180}^h 4e^{0.013h} \, dh$$

$$qx = \frac{-4}{0.013} [e^{0.013h} - e^{0.013(-180)}]$$

$$x = \frac{-307.7e^{0.013h} + 29.6}{-5.56}$$

The following figure shows $h(x)$:

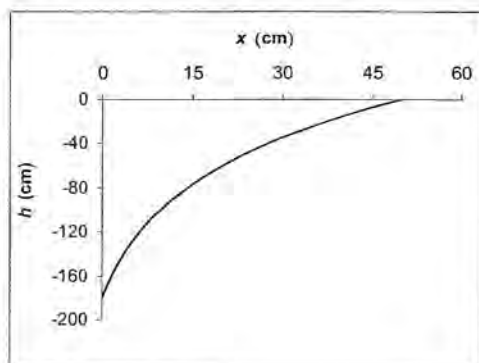


Fig. 4-7: Pressure head distribution

Problem 4-6

A 100 cm long vertical soil column has a steady state water flow situation. The top of the soil column is open to atmosphere. A water table is maintained at the bottom of the soil column. The saturated hydraulic conductivity, K_s is 0.05 m d^{-1} , and the pressure head, h at 20 cm above the water table ($z = 0.2 \text{ m}$) is -0.3 m . The air-entry value, h_{ae} of the soil is -0.35 m .

Assume that the reference level is at the bottom of the soil column where z is 0 and that z is positive upward. A positive q indicates that flow of water is in the upward direction.

a. Compute the steady state evaporation rate.

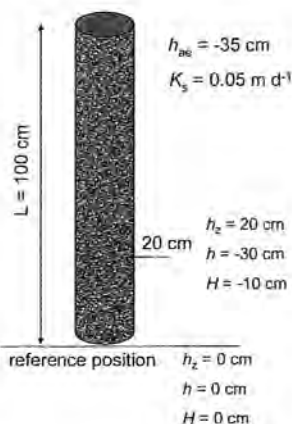


Fig. 4-8: Vertical soil column under steady state evaporation rate

Answer:

$$q = -K_s \left(\frac{dH}{dz} \right) = -0.05 \frac{(-0.3 + 0.2) - (0 + 0)}{0.2 - 0} = 0.025 \text{ m d}^{-1}$$

Therefore, 0.025 m d^{-1} is the steady state evaporation rate.

b. Calculate the height of the saturated zone for this situation.

Answer:

To calculate the height of the saturated zone, set $h = h_{ae}$ in Darcy's equation and solve for h_z .

$$0.025 = -0.05 \frac{(-0.35 + h_z) - (0 + 0)}{h_z - 0}$$

$$-0.5 = \frac{-0.35 + h_z}{h_z}$$

$$-0.5 h_z = -0.35 + h_z$$

$$h_z = \frac{-0.35}{-1.5} = 0.233 \text{ m}$$

which occurs at $z = h_z$.

c. Does this height remain constant if the evaporative rate is decreased?

Answer:

No, try setting the evaporative rate = $0.025/2$ and re-solve for h_z as in 4-6b

$$\frac{0.025}{2} = -0.05 \frac{(-0.35 + h_z) - (0 + 0)}{h_z - 0}$$

$$-0.25 = \frac{-0.35 + h_z}{h_z}$$

$$-0.25 h_z = -0.35 + h_z$$

$$h_z = \frac{-0.35}{-1.25} = 0.28 \text{ m}$$

We can see that as q decreases h_z increases.

d. Plot hydraulic head, gravitational head, and pressure head through the soil column for 4-6b.

Answer:

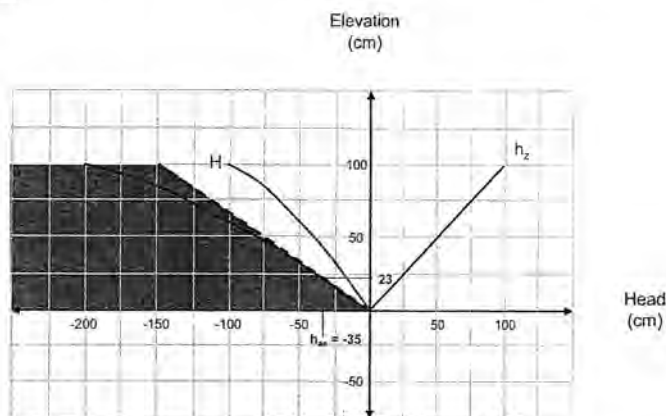


Fig. 4-9: Hydraulic, gravitational and pressure head distribution

In order for upward movement of water to occur, the pressure head (h) line must be in the shaded area starting from a pressure head value of -35 cm at an elevation of 23 cm, and decreasing over the length of the column. Therefore, the hydraulic head (H) will decrease as you move up through the column.

If the pressure head were along the dashed line, there would have to be a constant hydraulic conductivity through the column, which is not possible as the system switches from saturated to unsaturated at the point where the air-entry pressure is found in the column, at 23.3 cm.

The solid example line is curved because as h decreases so does K ; therefore, the slope of the H (and h) line (hydraulic head gradient) must change with height to maintain steady flow of water throughout the column.

Problem 4-7

A rain storm of intensity (R) 0.5 cm h^{-1} began at midnight on a sloping soil. The soil's saturated hydraulic conductivity, K_s is 0.1 cm h^{-1} and sorptivity, S is $0.3 \text{ cm h}^{-1/2}$. Estimate the time at which runoff commenced during the storm event.

Answer:

A formula to estimate the time to ponding (t_p) of water (or initiation of runoff) on the soil surface when rainfall rate, R , exceeds K_s can be

derived from the Green and Ampt (1911) equation for infiltration accordingly,

$$i = K_s + \frac{S^2}{2I}$$

where $i = R$, and $I = it_p$

Thus,

$$R = K_s + \frac{S^2}{2Rt_p}$$

$$t_p = \frac{S^2}{2R(R - K_s)} = \frac{0.3^2}{2(0.5)(0.5 - 0.1)} = 0.225 \text{ h}$$

Runoff would have begun at 12:13:30 am.

Note: This equation for estimating t_p is valid only for $R > K_s$.

Problem 4-8

A farmer wishes to irrigate a lettuce crop to a depth, L of 20 cm. Calculate the time required to irrigate the soil under a very thin (essentially zero) ponded depth of water at the soil surface if the saturated hydraulic conductivity of the soil, $K_s = 2 \text{ cm h}^{-1}$, the saturated and initial soil volumetric water contents are $\theta_s = 0.35$ and $\theta_i = 0.1$, respectively and the wetting front pressure head of the soil is, $h_f = -30 \text{ cm}$.

Answer:

Sufficient information is given in the question to allow use of the Green and Ampt (1911) model of infiltration. Actually, this model is ideally suited to this application, which requires irrigating to a specified, sharp, depth below the soil surface. The specified depth could, for example, be a crop-rooting zone.

$$t = \frac{(\theta_s - \theta_i)}{K_s} \left[L + h_f \ln \left(1 + \frac{L}{-h_f} \right) \right] = \frac{(0.35 - 0.1)}{2} \left[20 + (-30) \ln \left(1 + \frac{20}{30} \right) \right] = 0.58 \text{ h}$$

Problem 4-9

Consider the horizontal soil column such as in Figure 4-6 of length $\Delta x = 50 \text{ cm}$. The pressure head at $x = 50 \text{ cm}$ is $h = -20 \text{ cm}$ and at $x = 0$ is $h = -50 \text{ cm}$.

a. What is the overall hydraulic gradient?

Answer:

$$\frac{dH}{dx} = \frac{dh}{dx} = \frac{-20 - (-50)}{(50 - 0)}$$

$$\frac{dH}{dx} = 0.6$$

b. Is the flow from "right to left" or "left to right"?

Answer:

Flow direction is from right to left, from high (-20 cm) to low (-50 cm) hydraulic head. Note that the flow is negative, i.e., from higher to lower values of x .

c. What is a reasonable representative value of the pressure head within the soil?

Answer:

$$h_{core} = \frac{(-20) + (-50)}{2} = -35 \text{ cm}$$

Based on the arithmetic average of pressure heads at each end of the column.

Problem 4-10

Suppose you are interested in determining the unsaturated hydraulic conductivity of a soil. You wish to apply either the tension infiltrometer or the pressure outflow method.

a. Discuss briefly the conditions which would favor the tension infiltrometer over the pressure outflow method.

Answer:

Tension infiltrometers are used to measure the *in situ* steady-state infiltration of water under negative pressure, into a soil surface. This method is used for obtaining a wetting $K(h)$ curve, and employs a porous plate by which infiltrating water is kept under suction before entering the soil surface. Thus, flow is through the soil matrix and not through macropores.

Advantages of the tension infiltrometer method for gathering unsaturated hydraulic conductivity data are that the method is meant for field conditions, relatively simple to run and analyze, and relatively short in

duration. Disadvantages include the potential for air entrapment, and assumptions inherent in the methodology, such as isotropic and steady-state conditions.

b. Discuss briefly the conditions which would favor the pressure outflow method over the tension infiltrometer.

Answer:

The pressure outflow method is a laboratory, transient flow method where water is introduced into the soil under positive pressure. This method can be used to measure either the wetting or drying $K(h)$ curve in a single experiment, which is an advantage over the tension disc infiltrometer. The system is also relatively simple to set up. One problem encountered in the application of this method is that of hydraulic resistance between the porous plate on which the sample is placed, and of the soil-to-plate contact zone. Another disadvantage is that the measurement period is lengthy.

Problem 4-11

Consider the two hydraulic conductivity curves shown below.

a. Which of the two curves represents a “sandy soil” and which one a “clayey soil”?

Answer:

Generally speaking, water flow through sand at near saturated conditions experiences less impedance than through clay, as sandy soil pores are larger, and water retention of the larger, quartz-dominated sand particles is less than in clayey soils (unless the clayey soil is cracked or contains numerous macropores). Thus, under saturated conditions, sand should have higher K_s than clay. As water pressures decrease, sandy soil should drain more readily, leaving discontinuous, patches of adsorbed water throughout the large soil pore space, which are essentially immobile. Thus, $K(h)$ curves for sand should display a wider range of K values with varying negative pressures, while the range of K values with changing h for clay should be smaller. So Soil 2 is the clayey soil while Soil 1 is the sandy soil.

b. Why is there an intersection of the two curves?

Answer:

Sandy soils have a greater K_s value than clayey soils (without macropores); however, since sandy soils drain more readily than clayey soils under small negative pressure heads, the two curves eventually

intersect, and the unsaturated hydraulic conductivity of sand drops below clay.

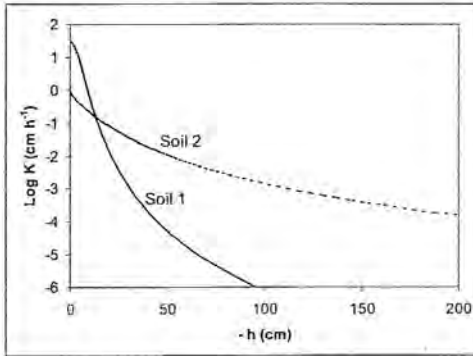


Fig. 4-10: Hydraulic conductivity functions of two soils

Problem 4-12

For horizontal flow, show how Darcy's law may be rewritten in terms of a water content gradient. What is the corresponding proportionality constant?

Answer:

$$q = -K(h) \left(\frac{dh}{dx} \right)$$

According to the chain rule, $\frac{dh}{dx} = \left(\frac{dh}{d\theta} \right) \left(\frac{d\theta}{dx} \right)$

Thus, $q = -K(h) \left(\frac{dh}{d\theta} \right) \left(\frac{d\theta}{dx} \right)$, where $\frac{dh}{d\theta}$ is the reciprocal of the specific water capacity ($1/C$)

Soil water diffusivity, D can be expressed as the relation between K and C , $D = K/C$

$$\text{We have } q = \left(\frac{-K(h)}{C} \right) \left(\frac{d\theta}{dx} \right) \rightarrow q = -D \left(\frac{d\theta}{dx} \right)$$

Problem 4-13

You have the following setup:

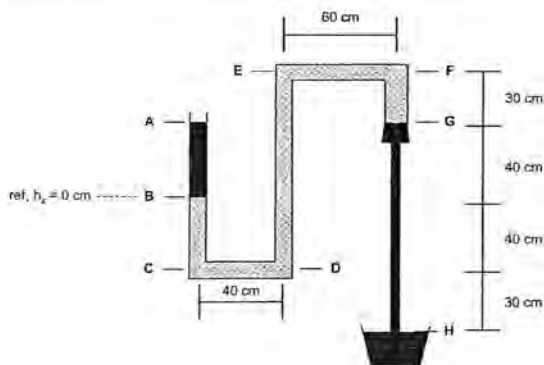


Fig. 4-11: Soil column

A to B is ponded water.

B to G is soil.

G to H is a water-filled tube.

Distance from C to D is 40 cm.

Distance from E to F is 60 cm.

Use point B as the reference level.

a. Assuming a steady state flux density, make a table of H , h_z , and h for points A through H.

Answer:

The total distance from point B to point G is

$$\Delta s = 40 + 40 + 110 + 60 + 30 = 280 \text{ cm}$$

The hydraulic head at position B is,

$$H_B = 0 + 40 = 40 \text{ cm}$$

The hydraulic head at position G is,

$$H_G = 40 + (-110) = -70 \text{ cm}$$

The total change in hydraulic head over this distance is,

$$H_G - H_B = -70 - 40 = -110 \text{ cm}$$

The hydraulic gradient over the entire soil column is,

$$\frac{\Delta H}{\Delta s} = \frac{-110}{280} = -0.393$$

Position A

$$h_z = 40 \text{ cm}, \quad h = 0 \text{ cm}, \quad H = 40 \text{ cm}$$

Position B

$$h_z = 0 \text{ cm}, \quad h = 40 \text{ cm}, \quad H = 40 \text{ cm}$$

Position C

$$h_z = -40 \text{ cm}$$

$$H = H_{\text{position B}} + \left[\left(\frac{\Delta H}{\Delta s} \right) (\text{total distance from position B}) \right]$$

$$H = 40 + [(-0.393)(40)] = 24.28 \text{ cm}$$

$$h = H - h_z = 24.28 - (-40) = 64.28 \text{ cm}$$

Position D

$$h_z = -40 \text{ cm}$$

$$H = H_{\text{position B}} + \left[\left(\frac{\Delta H}{\Delta s} \right) (\text{total distance from position B}) \right]$$

$$H = 40 + [(-0.393)(80)] = 8.56 \text{ cm}$$

$$h = H - h_z = 8.56 - (-40) = 48.56 \text{ cm}$$

Position E

$$h_z = 70 \text{ cm}$$

$$H = H_{\text{position B}} + \left[\left(\frac{\Delta H}{\Delta s} \right) (\text{total distance from position B}) \right]$$

$$H = 40 + [(-0.393)(190)] = -34.67 \text{ cm}$$

$$h = H - h_z = -34.67 - 70 = -104.67 \text{ cm}$$

Position F

$$h_z = 70 \text{ cm}$$

$$H = H_{\text{position B}} + \left[\left(\frac{\Delta H}{\Delta s} \right) (\text{total distance from position B}) \right]$$

$$H = 40 + [(-0.393)(250)] = -58.25 \text{ cm}$$

$$h = H - h_z = -58.25 - 70 = -128.25 \text{ cm}$$

Position G

$$h_z = 40 \text{ cm}$$

$$H = H_{\text{position B}} + \left[\left(\frac{\Delta H}{\Delta s} \right) (\text{total distance from position B}) \right]$$

$$H = 40 + [(-0.393)(280)] = -70 \text{ cm}$$

$$h = H - h_z = -70 - 40 = -110 \text{ cm}$$

Position H

$$h_z = -70 \text{ cm}, h = 0 \text{ cm}, H = -70 \text{ cm}$$

Summary:

Table 4-2: Heads at various points

	$h \text{ (cm)}$	$h_z \text{ (cm)}$	$H \text{ (cm)}$
A	0	40	40
B	40	0	40
C	64.28	-40	24.28
D	48.56	-40	8.56
E	-104.67	70	-34.67
F	-128.25	70	-58.25
G	-110	40	-70
H	0	-70	-70

b. If air leaked into the system at the rubber stopper at G, what additional information would you need (and how would you obtain it) to determine $K(\theta)$ between E and F?

Answer:

In order to determine the $K(\theta)$, the steady state volumetric flow rate, Q at G would have to be measured using a stopwatch and graduated cylinder, for example. If air leaked into the system, the calculated pressure heads listed above for points E and F would not be accurate, and thus measurements of the pressure heads and the average volumetric water content, θ , between E and F would also be required to determine $K(\theta)$.

Problem 4-14

A double-ring infiltration test was run in the field. Given the data below, calculate the infiltration rate, i over the time period given. Final units should be expressed in cm h^{-1} . Assume area of inner ring is 700 cm^2 .

Table 4-3: Water volume in supply reservoir

Actual Time	Water Volume in Supply Reservoir (cm ³)
9:00 AM	4000
9:25 AM	1500

Answer:

$$Q = \frac{V}{t} = \frac{2500 \text{ cm}^3}{25 \text{ min}} = \frac{100 \text{ cm}^3}{\text{min}} = 6000 \text{ cm}^3 \text{ h}^{-1}$$

$$i = \frac{Q}{A} = \frac{6000 \text{ cm}^3 \text{ h}^{-1}}{700 \text{ cm}^2} = 8.6 \text{ cm h}^{-1}$$

Problem 4-15

Compute and plot the soil water diffusivity, $D(\theta)$, and the specific water capacity, $C(\theta)$, functions for Sarpy loam soil characterized by the following van Genuchten parameters:

$$\alpha_{VG} = 2.79 \text{ m}^{-1} \quad n = 1.6 \quad \theta_s = 0.4 \text{ m}^3 \text{ m}^{-3} \quad \theta_r = 0.03 \text{ m}^3 \text{ m}^{-3} \quad K_s = 50 \text{ mm h}^{-1}$$

Answer:

The specific water capacity, $C(\theta)$, is given by

$$C(\theta) = \frac{d\theta}{dh}$$

and the soil water diffusivity, $D(\theta)$, by

$$D(\theta) = \frac{K(\theta)}{C(\theta)}$$

The analytical form of $C(\theta)$ for the van Genuchten (1980) model is

$$C(\theta) = \frac{d\theta}{dh} = \frac{(\theta_s - \theta_r)(n-1)\alpha_{VG}^n |h|^{n-1}}{\left[1 + (\alpha_{VG} |h|)^n\right]^{\left(2 - \frac{1}{n}\right)}}$$

Use the van Genuchten (1980) expression based on the Mualem (1976) model, for the hydraulic conductivity as a function of soil water content,

$$K(\theta) = K_s \left(\frac{\theta - \theta_r}{\theta_s - \theta_r} \right)^{1/2} \left[1 - \left(1 - \left(\frac{\theta - \theta_r}{\theta_s - \theta_r} \right)^{n/(n-1)} \right)^{1-(1/n)} \right]^2$$

See EXCEL file for numerical calculation resulting in the following plot:

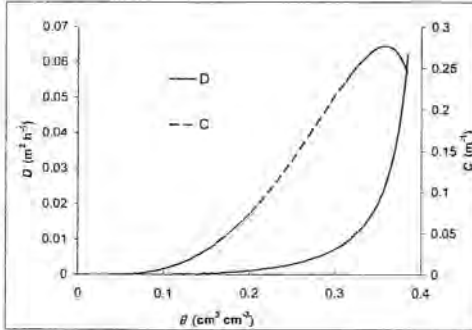


Fig. 4-12: Soil water diffusivity and the specific water capacity for Sarpy Loam soil

Problem 4-16

What is the equivalent depth (mm) of plant available soil water stored in the top 1.3 m of a soil characterized by the following Brooks-Corey (1964) model?

$$\theta = (\theta_s - \theta_r) (\alpha_{BC} |h|)^{-\lambda} + \theta_r$$

Given:

$\alpha_{BC} = 1.00 \text{ m}^{-1}$, $\lambda = 0.45$, $\theta_r = 0.02 \text{ m}^3 \text{ m}^{-3}$, $\rho_b = 1300 \text{ kg m}^{-3}$ and $\rho_s = 2650 \text{ kg m}^{-3}$

Answer:

The saturated volumetric water content is equal to the porosity (ϕ):

$$\theta_s = \phi = 1 - \frac{\rho_b}{\rho_s} = 1 - \frac{1300}{2650} = 0.51 \text{ m}^3 \text{ m}^{-3}$$

PAW is the Plant Available Water, which is determined by,

$$PAW = \theta_c - \theta_{wp}$$

Assume soil water pressure head at field capacity is

$$h_{fc} = -300 \text{ cm} = -3 \text{ m}$$

Assume soil water pressure head at wilting point is

$$h_{wp} = -15000 \text{ cm} = -150 \text{ m}$$

Substituting to determine the volumetric water content at field capacity,

$$\theta_{fc} = (0.51 - 0.02) |(-1)(-3)|^{-0.45} + 0.02 = 0.32 \text{ m}^3 \text{ m}^{-3}$$

$$\theta_{wp} = (0.51 - 0.02) |(-1)(-150)|^{-0.45} + 0.02 = 0.07 \text{ m}^3 \text{ m}^{-3}$$

Now determine PAW for the top 1.3 m of the soil profile:

$$PAW = \theta_{fc} - \theta_{wp} = 0.32 - 0.07 = 0.25 \text{ m}^3 \text{ m}^{-3}$$

$$PAW \text{ over 1.3 m soil is: } (0.25)(1.3) = 0.325 \text{ m or 325 mm}$$

Problem 4-17

Given the $h(\theta)$ and $K(\theta)$ curves shown below, label each appropriately and estimate the soil water capacity (C) and soil water diffusivity (D) at $\theta = 0.3$. Furthermore, estimate α_G from Gardner's expression for the unsaturated hydraulic conductivity as a function of the pressure head at this same water content.

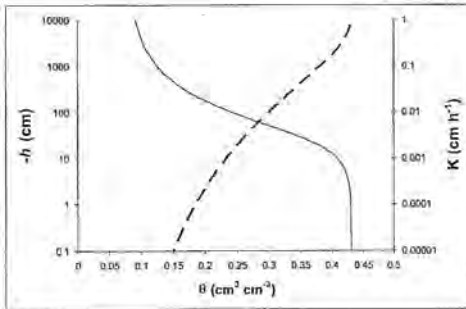


Fig. 4-13: Soil hydraulic functions

Answer:

The dashed line is hydraulic conductivity, K , and the solid line is the pressure head, h .

The soil water capacity, C , is essentially the slope of the $\theta(h)$ curve. To determine the slope at $\theta = 0.3 \text{ m}^3 \text{ m}^{-3}$, choose two points on either side of this point and calculate the slope using the corresponding values on the x and y axes:

$$C(\theta) = \frac{d\theta}{dh} = \frac{0.35 - 0.25}{-28 - (-90)} = 0.00161 \text{ cm}^{-1}$$

The soil water diffusivity, D , is the ratio of K and C :

$$D(\theta) = \frac{K(\theta)}{C(\theta)}$$

The hydraulic conductivity at $\theta = 0.3 \text{ m}^3 \text{ m}^{-3}$ is $\sim 0.01 \text{ cm h}^{-1}$ leading to

$$D(\theta) = \frac{K(\theta)}{C(\theta)} = \frac{0.01 \text{ cm h}^{-1}}{0.00161 \text{ cm}^{-1}} = 6.21 \text{ cm}^2 \text{ h}^{-1}$$

Gardner's Equation is

$K(h) = K_s e^{\alpha_G h}$ and the above K can be used to find α_G :

$$0.01 \text{ cm h}^{-1} = 1 \text{ cm h}^{-1} e^{\alpha_G (-50 \text{ cm})}$$

$$\alpha_G = 0.092 \text{ cm}^{-1}$$

Problem 4-18

a. In terms of frequency in nature, dominant water potential components, and hydraulic conductivity, compare and contrast saturated and unsaturated water-flow conditions.

Answer:

Most of the processes involving soil-water interactions in the field occur in unsaturated soils. Plant water and nutrient uptake and water and solute transport below the root zone are soil processes commonly occurring under unsaturated conditions. Saturated conditions are generally found below a water table, either as part of the permanent groundwater system or in the vadose zone as a perched aquifer. While under both saturated and unsaturated conditions, soil water flow is from increasing to decreasing hydraulic head, the hydraulic head components and structure of the flow matrix alternates between the two states. For example, under saturated conditions, the forces leading to water flow are gravity and the positive pressure head, while in unsaturated soils, the dominating hydraulic head component other than gravity is the negative pressure head or suction. Another soil property significantly affecting water flow under saturated and unsaturated conditions is the hydraulic conductivity. The flow rate through saturated soils is maximal, as all pores participate in the conducting process, and water remains in an unbroken, continuous column throughout pores. Under drying conditions, as small negative pressure heads are applied to soils, the hydraulic conductivity drops sharply, slowing the rate of water transfer through soil. Generally speaking, solving flow problems for unsaturated soil conditions according to the continuity equation (Richards, 1931),

$$\frac{\partial \theta}{\partial t} = -\frac{\partial}{\partial z} \left[K \left(1 + \frac{\partial h}{\partial z} \right) \right]$$

often requires the use of various pedo-transfer functions based on the soil water potential. Analytical solutions to the water flow equation are complicated by the fact that the unsaturated hydraulic conductivity is a non-linear function of water content, for example Mualem (1976):

$$K(\theta) = K_s \left(\frac{\theta - \theta_r}{\theta_s - \theta_r} \right)^{1/2} \left[1 - \left(1 - \left(\frac{\theta - \theta_r}{\theta_s - \theta_r} \right)^{n/(n-1)} \right)^{1-(1/n)} \right]^2$$

and the soil water content is a non-linear function of the soil water potential, for example van Genuchten (1980):

$$\theta = (\theta_s - \theta_r) \left[1 + \alpha_{VG} |h|^n \right]^{((1/n)-1)} + \theta_r$$

Under saturated and incompressible conditions, no change in the soil water content is expected, simplifying the continuity equation,

$$\frac{\partial}{\partial z} \left[K \left(1 + \frac{\partial h}{\partial z} \right) \right] = 0$$

b. Consider an unsaturated soil column composed of a loamy sand layer and a silt loam layer of equal thickness. Describe what happens to water under steady flow conditions as it moves vertically downward from the top of the column and encounters the interface (i.e., textural discontinuity) between the two soil layers when 1) the loamy sand overlies the silt loam, and 2) the silt loam overlies the loamy sand. See Figure 5-19 in Warrick (2003) for an example to follow.

Answer:

Let us assume unsaturated, dry conditions throughout the soil column, with 10 cm of ponded water at the top. As water infiltrates through the profile, the hydraulic conductivities of the two soils are expected to rise—the loamy sand more drastically than the silt loam, as the range of $K(h)$ values for sandy soils is larger than for silty-clayey soils, as depicted in Figure 4-10. Thus, under conditions of free drainage, the pressure head at the top of the loamy sand layer is expected to be smaller ('more negative') than at the bottom, while the pressure head at the top of the silt loam layer is expected to be larger ('more positive') than at the bottom of the layer.

For the first scenario, where loamy sand overlies silt loam, the pressure at the top of the profile would be positive, increase until the top of the silt-loam layer, and then gradually decrease with depth, until settling at atmospheric pressure at the bottom of the column. Infiltration into the coarser-textured soil would be quicker than into the finer-textured soil, which would lead to a buildup of positive pressure above the silt loam layer.

For the second scenario, where silt loam overlies loamy sand, the pressure at the top of the profile would sharply decrease with depth due to reduced conductivity rates. The pressure gradient would then reverse at the interface between the two soils, where conductivity rates would rise

sharply, and the pressure is expected to increase until arriving at atmospheric pressures at the bottom of the loamy sand layer.

c. Suppose you have a column of composite soil with a clayey horizon overlying a sandy horizon. Under saturated, steady-state flow conditions in the clayey horizon, is the sandy horizon below saturated? Explain your response.

Answer:

This scenario can be likened to that of a disc-infiltrometer used for measuring the unsaturated hydraulic conductivity in soils. Since the saturated hydraulic conductivity of clay is lower than most other soil types, and surely lower than coarse-textured sand, infiltration into the sandy soil would be at a rate less than the saturated hydraulic conductivity. Thus, so long as the clay layer remained saturated and flow conditions were steady-state, the sandy soil would never fully saturate, and the hydraulic conductivity would remain at less than its maximal rate.

d. A silt-loam soil has a B_{hs} horizon that is 12 cm thick. Water is moving into the top of the horizon at a flux density of $q = -0.32 \text{ cm h}^{-1}$ and out of the horizon at a flux density of $q = -0.44 \text{ cm h}^{-1}$ (we are continuing to assume the z -axis is chosen positive upwards; hence a downward flux component will be negative). Calculate the approximate time rate of change of the volumetric soil water content in the horizon. Is the horizon gaining or losing water?

Answer:

Start with continuity equation

$$\frac{\partial \theta}{\partial t} = -\frac{\partial q}{\partial z}$$

We are given the change in flux over the horizon,

$$\frac{\partial q}{\partial z} = \frac{q_{\text{top}} - q_{\text{bot}}}{z_{\text{top}} - z_{\text{bot}}} = \frac{-0.32 - (-0.44)}{12 - 0} = 0.014 \text{ h}^{-1}$$

Thus, the change in water content over time is:

$$\frac{\partial \theta}{\partial t} = -0.01 \text{ h}^{-1}$$

and the horizon is losing water.

e. A composite soil column 100-cm long was created composed of two soil layers. The top layer has a thickness of 70 cm and a K_s of $2.3 (10)^{-5} \text{ m s}^{-1}$. The bottom horizon has a thickness of 30 cm and a K_s of $1.6 (10)^{-8} \text{ m s}^{-1}$. Assuming that a constant head of 10 cm of water exists at the soil

surface, calculate i) the effective hydraulic conductivity (K_s^*), ii) steady-state flux density of water through the column, and iii) the pressure head at the interface between the two layers.

Answer:

i)

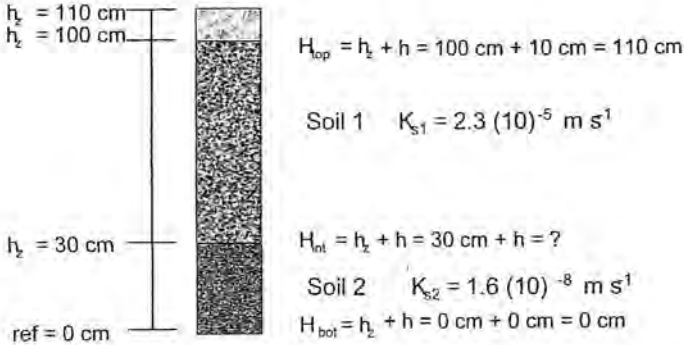


Fig. 4-14: Water flow in a column with two soils

$$K_s^* = \frac{s_1 + s_2}{\frac{s_1}{K_{s1}} + \frac{s_2}{K_{s2}}} = \frac{70 + 30}{\frac{70}{2.3(10)^{-3}} + \frac{30}{1.6(10)^{-6}}} = 5.325(10)^{-6} \text{ cm s}^{-1} \text{ or } 5.325(10)^{-8} \text{ m s}^{-1}$$

ii) assume z is positive upward. A negative q indicates that flow of water is in the downward direction.

$$q = -K_s^* \frac{dH}{dz} = -K_s^* \frac{(H_{\text{top}} - H_{\text{bot}})}{(z_{\text{top}} - z_{\text{bot}})} = -5.325(10)^{-6} \frac{(110 - 0)}{(100 - 0)} = -5.857(10)^{-6} \text{ cm s}^{-1}$$

iii) Since the flux is at steady-state, start by equating the flux through the two layers:

$$-K_{s1} \frac{dH}{dz_1} = -K_{s2} \frac{dH}{dz_2}$$

$$K_{s1} \frac{(H_{\text{top}} - H_{\text{int}})}{(z_{\text{top}} - z_{\text{int}})} = K_{s2} \frac{(H_{\text{int}} - H_{\text{bot}})}{(z_{\text{int}} - z_{\text{bot}})}$$

$$H_{\text{int}} = \frac{3.615(10)^{-3}}{3.291(10)^{-5}} = 109.85 \text{ cm}$$

Now solve for the soil water pressure head at the interface,

$$H_{\text{int}} = h_{z_{\text{int}}} + h_{\text{int}} \rightarrow h_{\text{int}} = H_{\text{int}} - h_{z_{\text{int}}}$$

$$h_{\text{int}} = 109.85 - 30 = 79.85 \text{ cm}$$

Another way to solve this is using the steady-state flux through the column used above:

$$q = -K_{s2} \frac{dH}{dz} \rightarrow -K_{s2} \frac{(H_{\text{int}} - H_{\text{bot}})}{(z_{\text{int}} - z_{\text{bot}})}$$

$$H_{\text{int}} = \frac{q}{-K_{s2}} (z_{\text{int}} - z_{\text{bot}}) + H_{\text{bot}}$$

$$H_{\text{int}} = \frac{-5.857(10)^{-6}}{-1.6(10)^{-6}} (30 - 0) + 0 = 109.82 \text{ cm}$$

$$h_{\text{int}} = H_{\text{int}} - h_{z_{\text{int}}} = 109.85 - 30 = 79.85 \text{ cm}$$

The pressure head at the interface is 79.85 cm. Note that it is important to express the answer to the nearest 0.01 cm in this case. If you round off the answer to 80 cm, H_{int} will equal 110 cm and the flux through the top layer is zero.

Problem 4-19

Shani et al. (1987) measured the steady state pond radii, r_0 under various volumetric flow rates, Q given in the following table. What are the saturated hydraulic conductivity (K_s) and the matric flux potential (ϕ_m) of the initially air-dried Hazerim loamy sand?

Table 4-4: Steady state pond radii for various volumetric flow rates

$Q \text{ (cm}^3 \text{ h}^{-1}\text{)}$	$r_0 \text{ (cm)}$
1000	1.7
2000	3.3
4000	6.4
8000	12.2

Use the Wooding (1968) equation for steady state flux from a surface disc

$$q = K_s + \frac{4}{\pi} \frac{K_s}{\alpha_G} \frac{1}{r_0}$$

Answer:

Assume that, $q = \frac{Q}{\pi r_0^2}$ and $\frac{K_s}{\alpha_G} = \phi_m$ where ϕ_m is the matric flux potential.

Regression on q versus r_0^{-1} yields $K_s = 2.36 \text{ cm h}^{-1}$ as the intercept, and $\frac{4}{\pi} \phi_m$ as the slope resulting in $\phi_m = \frac{183.59\pi}{4} = 144.19 \text{ cm}^2 \text{ h}^{-1}$

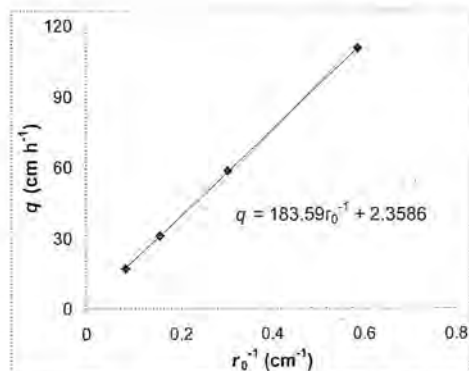


Fig. 4-15: Infiltration rates as a function of the reciprocal of disc radius

Problem 4-20

Consider a 1 m long horizontal column with homogenous air-dried loamy soil. A constant pressure head of 0.5 m was applied to the left side of the column. The wetting front was at 0.6 m after 10 h. After fully saturating the soil, a 50 cm^3 volume of water was collected after 2 additional hours. The diameter of the column is 4 cm, and the saturated and residual water contents are 0.43 and 0.08, respectively. The horizontal coordinate x is zero on the left and increases to the right.

a. What is the wetting front pressure head of this soil?

Answer:

To answer this question use the relationships for sorptivity S :

$$I = S t^{\frac{1}{2}}$$

$$S = 2 \sqrt{\frac{(\theta_{\text{wet}} - \theta_{\text{dry}}) K_s (0 - h_f)}{\pi}}$$

We solve for the sorptivity S , assuming that as infiltration proceeds, wetted areas are fully saturated:

$$I = St^{0.5} \rightarrow S = \frac{I}{t^{0.5}}$$

$$S = \frac{60(0.43 - 0.08)}{(10)^{0.5}} = 6.641 \text{ cm h}^{-0.5}$$

Now solve for the saturated hydraulic conductivity from the saturated data. Start by determining the flux density.

$$q = \frac{Q}{A}$$

$$Q = \frac{50 \text{ cm}^3}{2 \text{ hr}} = 25 \text{ cm}^3 \text{ h}^{-1}$$

$$A = \pi \left(\frac{4 \text{ cm}}{2} \right)^2 = 12.566 \text{ cm}^2$$

$$q = \frac{25 \text{ cm}^3 \text{ h}^{-1}}{12.566 \text{ cm}^2} = 1.989 \text{ cm h}^{-1}$$

Use the flux density to determine the saturated hydraulic conductivity:

$$q = -K_s \frac{\Delta H}{\Delta x} \rightarrow K_s = -\frac{q}{\frac{\Delta H}{\Delta x}} = -\frac{1.989}{\frac{(0-50)}{(100-0)}} = 3.979 \text{ cm h}^{-1}$$

Finally, rearrange the expression for sorptivity to determine the wetting front pressure head, h_f

$$S = 2\sqrt{\frac{(\theta_{\text{wet}} - \theta_{\text{dry}}) K_s (0 - h_f)}{\pi}}$$

$$h_f = -\frac{S^2 \pi}{4(\theta_{\text{wet}} - \theta_{\text{dry}}) K_s} = -\frac{(6.641)^2 \pi}{4(0.43 - 0.08) 3.979} = -24.87 \text{ cm}$$

b. How long did it take to fully saturate this column?

Answer:

$$I = St^{0.5} = L\Delta\theta$$

$$100(0.43 - 0.08) = 6.641t^{0.5}$$

$$t^{0.5} = \frac{100(0.43 - 0.08)}{6.641}$$

$$t^{0.5} = 5.27$$

$$t = 27.78 \text{ h}$$

c. Plot the infiltration rate, i as a function of time for this soil.

Answer:

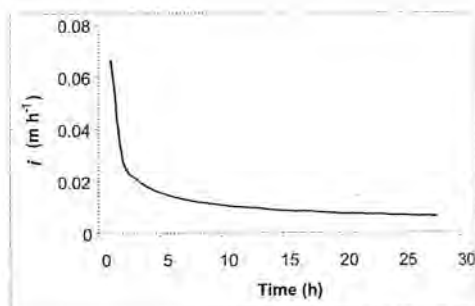


Fig. 4-16: Infiltration rate as a function of time

Problem 4-21

Consider a 1 m long horizontal column with homogenous loamy sand soil. Use HYDRUS-1D (Simunek et al., 2008) to plot the wetting front position as a function of different fixed (ranging from 0-10 cm) pressure heads at one end of the column at an elapsed time of 5 hours. Note that each run of HYDRUS-1D will have a constant pressure head at one end of the column. Assume that the initial pressure head throughout the soil is -500 cm. Repeat calculations with loamy soil with initial pressure head of -1000 cm.

Make the calculations using HYDRUS-1D with the following conditions:

- One-dimensional horizontal soil profile 1 m length.
- Duration of simulation: 5 hours
- Soil material: Loamy sand or loamy
- Discretization: 101 nodes equally distributed with depth (i.e. 1 cm per element)
- Initial condition: Pressure head is -500 cm or -1000 cm.

- "Lower" boundary condition: Constant pressure head of -500 cm or -1000 cm.
- "Upper" boundary condition: Constant pressure head that vary between 0 and 10 cm in increments of 2 cm.

Answer:

HYDRUS-1D may be downloaded free of charge at http://www.pc-progress.cz/Pg_Hydrus_1D.htm. Make up constant boundary conditions between 0 and 10 cm in HYDRUS-1D, and then from the results of the water content distribution determine where the wetting front is approximately located. The model outputs in the following figure were generated using six different simulations for each soil. The van Genuchten-Mualem single-porosity hydraulic model with no hysteresis was assumed.

(See attached HYDRUS-1D modules 4_21_LS_A-F.h1d and 4_21_L_A-F.h1d.)

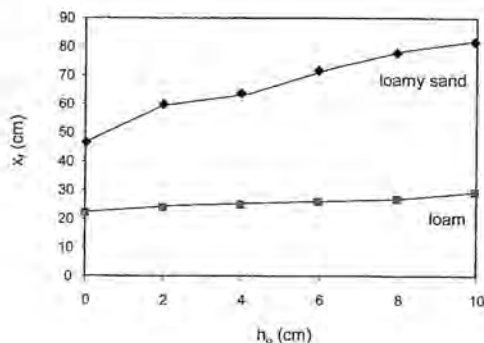


Fig. 4-17: Wetting front as a function of boundary pressure head in two soils

Problem 4-22

Use the RETC (van Genuchten et al., 1991) program to calculate the soil hydraulic properties of the van Genuchten hydraulic model (1980) for the following three sets of data (note that h is the same for each set). Plot the observed and predicted retention curves using $m = 1 - n^{-1}$. Discuss the differences between the results.

Table 4-5: Water contents as a function of pressure heads for 3 soils

$-h$ (cm)	θ_1 (cm ³ cm ⁻³)	θ_2 (cm ³ cm ⁻³)	θ_3 (cm ³ cm ⁻³)
0	0.46	0.430	0.380
45.58	0.45	0.311	0.373
96.9	0.42	0.245	0.366
150	0.38	0.212	0.360
204	0.34	0.191	0.355
266.7	0.3	0.176	0.351
346.4	0.26	0.163	0.346
458.3	0.22	0.151	0.341
635.9	0.18	0.139	0.334
979.8	0.14	0.126	0.325
1990	0.1	0.110	0.310
7998	0.07	0.093	0.282

The RETC program may be downloaded free of charge at
http://www.pc-progress.cz/Pg_RetC.htm.

Answer:

Using the data given in the table, the soil hydraulic properties of the van Genuchten ($m = 1 - n^{-1}$) model for the three soils are given in the table below

Table 4-6: van Genuchten parameters for 3 soils

	Soil 1	Soil 2	Soil 3
θ_1 (cm ³ cm ⁻³)	0.06	0.078	0.061
θ_s (cm ³ cm ⁻³)	0.46	0.43	0.38
α_{VG} (cm ⁻¹)	0.005	0.036	0.008
n (-)	2.0	1.56	1.09

The RETC program produced the three best-fit retention curves below (note that the pressure head in RETC program is absolute value). Determine with, RETC, the soil hydraulic properties for the other seven models.

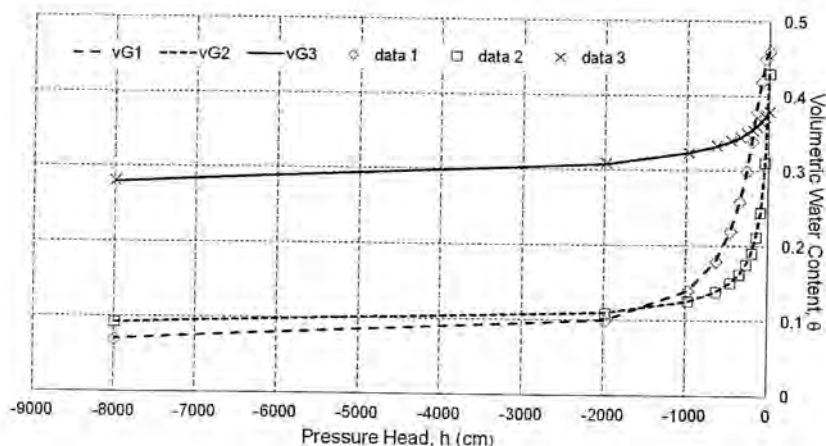


Fig. 4-18: Water retention data for the three soils with best-fit van Genuchten equation

Problem 4-23

a. Use ROSETTA (Schaap et al., 1998) to calculate the hydraulic parameters for the van Genuchten (1980) hydraulic model of the following particle sizes and bulk densities:

Table 4-7: Particle sizes and bulk densities for 4 soils

Soil #	Sand (%)	Silt (%)	ρ_b (g cm ⁻³)
1	20	70	1.2
2	40	50	1.3
3	60	20	1.4
4	80	5	1.5

The ROSETTA program can be downloaded free of charge from the US Salinity Laboratory's web site (<http://www.ars.usda.gov/Services/docs.htm?docid=8953>).

Answer:

The following table gives the van Genuchten model parameters for the four soils.

Table 4-8: van Genuchten parameters for 4 soils

Soil	α_{VG} (cm^{-1})	n (-)	θ_s ($\text{cm}^3 \text{cm}^{-3}$)	θ_r ($\text{cm}^3 \text{cm}^{-3}$)
1	0.004	1.7	0.44	0.06
2	0.007	1.6	0.39	0.05
3	0.02	1.4	0.42	0.06
4	0.03	1.6	0.40	0.06

b. What will be the corresponding parameters for the Brooks and Corey (1964) hydraulic model?

Answer:

The Brooks and Corey water retention model is

$$\theta = (\theta_s - \theta_r) (\alpha_{BC} |h|)^{-\lambda} + \theta_r \quad \text{for } \alpha_{BC} |h| > 1$$

To find the values of α_{BC} and λ for each soil listed in the table, use the van Genuchten parameters and the following van Genuchten water retention model to predict $\theta(h)$. Then use the predicted $\theta(h)$ in RETC to calculate the corresponding best-fit α_{BC} and λ values for each soil. For the first soil listed in the table, $\alpha_{BC} = 0.007 \text{ cm}^{-1}$ and $\lambda = 0.462$. van Genuchten model:

$$\theta = \theta_r + (\theta_s - \theta_r) \left(\frac{1}{1 + (\alpha_{VG} |h|)^n} \right)^m$$

$$\text{with } m = 1 - \frac{1}{n}$$

As visible in the following figure, the two models give nearly identical water retention curves using this approach except near saturation where the Brooks-Corey model reaches a plateau of saturation when the product of α_{BC} and absolute value of h exceeds one or when h exceeds the soil's air entry pressure head of -100 cm.

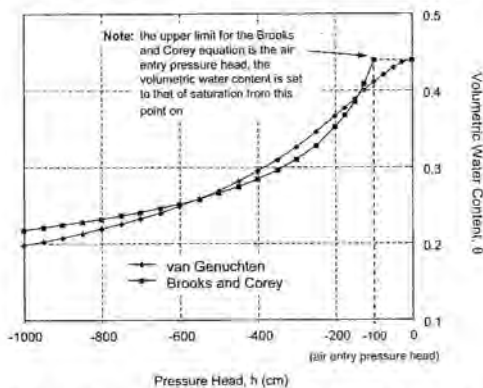


Fig. 4-19: Retention curves for two hydraulic models

Problem 4-24

Estimate the unsaturated hydraulic conductivity by the instantaneous profile method. Use the following TDR and tensiometer measurements measured at 10 cm and 30 cm below the soil surface. For further information see Vachaud and Dane (2002).

Table 4-9: Water content and pressure head measurements

z = 10 cm			z = 30 cm		
Time (h)	h (cm)	θ (cm ³ cm ⁻³)	Time (h)	h (cm)	θ (cm ³ cm ⁻³)
0.5	-13.7	0.392	0.5	-25.7	0.46
1	-15.2	0.38	1	-32.1	0.338
1.5	-43.2	0.311	1.5	-38.2	0.324
3.5	-52.1	0.296	3.5	-54.1	0.294
5.5	-60.1	0.284	5.5	-63.9	0.278
7.5	-66.3	0.275	7.5	-76.3	0.267
9.5	-82.5	0.260	9.5	-82.5	0.265

Answer

Define the loss in storage per unit area between $z = 0$ to $z = 20$ cm to be ΔS over a time change Δt . As there is no flow across the soil surface, the loss in storage is equal to the flux density across the 20 cm depth multiplied by Δt . Thus, by Darcy's law we have the approximate relationship for any time change:

$$\Delta S \approx -K \frac{\Delta H}{\Delta z} \Delta t$$

(We consider the profile to the 20 cm depth to match midway between the tensiometer depths). The average water contents are computed for each measurement time in the table below and the stored water found by multiplying by the profile depth 20 cm (we could average θ differently). The loss in storage ΔS follows in the next column for each time increment. For example, for 0.5 to 1 h the result is $\Delta S = 8.52 - 7.18 = 1.34$ cm. The next column is the hydraulic head at $z = 10$ cm (the choice of z as a depth implies the z -axis is positive downwards and the gravimetric head $h_z = -z$ with the reference level at the surface). Thus, the hydraulic head gradient at $t = 0.5$ is $[-55.7 - (-23.7)]/20 = -1.6$. The average hydraulic head for the first 2 times (-1.7225) is used in the above equation giving for the first time increment: $1.34 \approx -K(-1.7225)(0.5)$ or $K = 1.556$ cm h⁻¹. This value of K corresponds to the average pressure heads for the 2 times and 2 depths used in the calculation [$h = (-13.7 - 15.2 - 25.7 - 32.1)/4 = -21.675$ cm] or alternatively to the corresponding

average of water contents. Results for the 6 time increments are plotted in Figure 4-20.

Table 4-10: The instantaneous profile method after Vachaud and Dane (2002)

t	$\bar{\theta}$	S	ΔS	H_{10}	H_{30}	$\frac{\Delta H}{\Delta z}$	$\frac{\overline{\Delta H}}{\Delta z}$	K	$\bar{h}$
h	$\text{cm}^3 \text{cm}^{-3}$	cm	cm	cm	cm	cm cm^{-1}	cm cm^{-1}	cm h^{-1}	cm
0.5	0.426	8.52		-23.7	-55.7	-1.6			
1	0.359	7.18	1.34	-25.2	-62.1	-1.845	-1.7225	1.556	-21.67
1.5	0.3175	6.35	0.83	-53.2	-68.2	-0.75	-1.2975	1.279	-32.17
3.5	0.295	5.9	0.45	-62.1	-84.1	-1.1	-0.925	0.243	-46.9
5.5	0.281	5.62	0.28	-70.1	-93.9	-1.19	-1.145	0.122	-57.55
7.5	0.271	5.42	0.2	-76.3	-106.3	-1.5	-1.345	0.074	-66.65
9.5	0.2625	5.25	0.17	-92.5	-112.5	-1	-1.25	0.068	-76.9

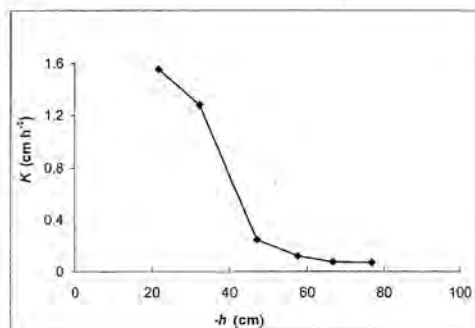


Fig. 4-20: Hydraulic conductivity versus soil-water pressure head

Problem 4-25

The following data was collected with a Guelph Pressure (Ring) Infiltrometer (GPI). Use the data and the equations given below to calculate the field-saturated hydraulic conductivity (K_{fs}) and the inverse macroscopic capillary length parameter (α_G) of the soil. The ring radius (r) is 5 cm and the depth of ring insertion (d) is 4 cm. Note that G is a dimensionless shape factor.

For more information on the GPI method see: (Reynolds and Elrick, 2002).

Table 4-11: Flow rates from GPI reservoir at different hydraulic heads

Steady volumetric flow rates from GPI reservoir		Applied hydraulic head at the soil surface	
Q_1	$= 35 \text{ cm}^3 \text{ min}^{-1}$	H_{A1}	$= 10 \text{ cm}$
Q_2	$= 42 \text{ cm}^3 \text{ min}^{-1}$	H_{A2}	$= 15 \text{ cm}$

Answer:

$$K_{fs} = \frac{\alpha_G G Q}{r(\alpha_G H_{A1} + 1 + G\alpha_G \pi r)}$$

$$G = 0.316 \frac{d}{r} + 0.184 = 0.316 \frac{4}{5} + 0.184 = 0.437$$

Since both sets of Q and H_A data were measured in the same plot, we can isolate α_G by equating two expressions for K_{fs} (two equations with two unknowns):

$$\frac{\alpha_G G Q_1}{r(\alpha_G H_{A1} + 1 + G\alpha_G \pi r)} = \frac{\alpha_G G Q_2}{r(\alpha_G H_{A2} + 1 + G\alpha_G \pi r)}$$

$$\frac{\alpha_G (0.437)(35)}{5(\alpha_G 10 + 1 + 0.437\alpha_G \pi 5)} = \frac{\alpha_G (0.437)(42)}{5(\alpha_G 15 + 1 + 0.437\alpha_G \pi 5)}$$

$$\alpha_G = 0.123 \text{ cm}^{-1}$$

Substituting α_G back into the original equation gives:

$$K_{fs} = \frac{(0.123)(0.437)(0.35)}{\{5[(0.123)(10) + 1 + (0.437)(0.123)\pi 5]\}} = 0.122 \text{ cm min}^{-1} = 2.033(10)^{-5} \text{ m s}^{-1}$$



Equations are more important to me, because politics is for the present, but an equation is something for eternity - Albert Einstein